

How long is a simulated double pendulum right?

Step size, floating-point precision, and the Lyapunov exponent

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Abstract

Students who simulate a chaotic system learn that nearby initial conditions separate exponentially, but they seldom ask the corresponding question about the simulation itself, which is perturbed at every step by truncation and by roundoff. We describe a computational exercise that answers it. The double pendulum is integrated with a fixed-step fourth-order Runge–Kutta method in arbitrary-precision arithmetic, so that the step h and the number of significant bits p can be varied independently, and we measure the time T for which each computed trajectory agrees with a far more accurate reference. The results follow a single formula in which the largest Lyapunov exponent λ is the only property of the pendulum that appears: halving h extends T by $4 \ln 2 / \lambda$, and each additional bit extends it by $\ln 2 / \lambda$, until the other source of error takes over. The numerical error therefore measures λ . The two values obtained this way, 0.35 ± 0.02 and 0.35 ± 0.01 in natural units, match the growth rate of a deliberately perturbed trajectory over the same interval and lie within 20% of the long-time exponent, 0.299. In double precision no choice of step keeps this pendulum accurate beyond about 101 natural time units, roughly 32 s for rods one meter long, and the computational work needed grows exponentially with T . Energy, meanwhile, remains conserved to a part in 10^{10} long after the trajectory has become wrong. The exercise needs about forty lines of code and turns numerical error from a nuisance into a measuring instrument.

I. INTRODUCTION

The double pendulum is the standard classroom example of chaos in a mechanical system, and this journal has treated it as one for thirty years. Shinbrot *et al.*[1] released two nearly identical physical pendulums together, watched them separate, and measured an exponential growth rate of $7.5 \pm 1.5 \text{ s}^{-1}$. Levien and Tan[2] built a double pendulum for the undergraduate laboratory and compared its motion with a numerical model. More recently D’Alessio[3] combined analysis, simulation and experiment and computed the full spectrum of Lyapunov exponents. All three ask how the *physical* system amplifies an uncertainty in its initial state.

A simulation is subject to the same amplification, with one difference. Its error is not confined to the initial condition. A small error is committed at every step, from truncating the Taylor series of the solution and from rounding each arithmetic result to a finite number of bits, and every one of those errors is amplified from the moment it is made. A student

who animates a double pendulum for a minute therefore has no way of knowing whether the last forty seconds bear any relation to the solution of the equations that were typed in. The checks students usually reach for do not settle the matter. Energy can be conserved to many digits by a trajectory that is entirely wrong, as we show in Sec. IV E, and halving the step and seeing no visible change tests only the interval over which nothing has yet gone wrong.

This paper describes an exercise that answers the question quantitatively. None of the underlying facts is new. Lorenz[4] found sensitive dependence by re-entering rounded numbers into a running computation; Lighthill[5] gave the horizon of predictability its name; the competition between truncation and roundoff is standard numerical analysis;[6] and the reliability of computed chaotic solutions has been studied in the research literature on atmospheric models[7] and with multiple-precision arithmetic.[8] What we have not found is a treatment at the level of an undergraduate course in which the student separates the two sources of error, measures each, and connects both to the Lyapunov exponent. That is what is offered here. The exercise teaches five things:

1. The time for which a simulation of a chaotic system is accurate depends only logarithmically on the error committed per step, so that large improvements in the integrator buy modest extensions of that time.
2. Truncation error and roundoff error can be isolated from one another cleanly once the precision of the arithmetic is a parameter of the program rather than a property of the hardware.
3. Numerical error is itself a small perturbation, and its growth measures the Lyapunov exponent as well as a deliberately perturbed initial condition does.
4. For a given floating-point format there is a best step, and a longest accurate time that no step can improve upon.
5. Conservation of energy is not evidence that a chaotic trajectory is correct.

II. MODEL AND NUMERICAL METHOD

A. Equations of motion

Two point masses m hang in series from a fixed pivot on massless rigid rods of length l and move in a vertical plane under gravity. With the angles θ_1 and θ_2 measured from the downward vertical and time in units of $\sqrt{l/g}$, the Lagrangian equations of motion contain no parameters:

$$\ddot{\theta}_1 = \frac{\omega_1^2 \sin \Delta \cos \Delta + \sin \theta_2 \cos \Delta + \omega_2^2 \sin \Delta - 2 \sin \theta_1}{2 - \cos^2 \Delta}, \quad (1)$$

$$\ddot{\theta}_2 = \frac{-\omega_2^2 \sin \Delta \cos \Delta + 2 (\sin \theta_1 \cos \Delta - \omega_1^2 \sin \Delta - \sin \theta_2)}{2 - \cos^2 \Delta}, \quad (2)$$

where $\omega_i = \dot{\theta}_i$ and $\Delta = \theta_2 - \theta_1$. The energy, in units of mgl , is

$$E = \omega_1^2 + \frac{1}{2}\omega_2^2 + \omega_1\omega_2 \cos \Delta - 2 \cos \theta_1 - \cos \theta_2. \quad (3)$$

Every run starts from rest with both rods horizontal, $\theta_1 = \theta_2 = \pi/2$, for which $E = 0$. One natural time unit is 0.319 s when $l = 1$ m.

B. Integrator and arithmetic

We write Eqs. (1) and (2) as a first-order system $\dot{\mathbf{y}} = \mathbf{f}(\mathbf{y})$ for $\mathbf{y} = (\theta_1, \theta_2, \omega_1, \omega_2)$ and advance it with the classical fourth-order Runge–Kutta method (RK4) at a fixed step h . A fixed step is essential to the exercise. An adaptive integrator adjusts h to hold its own estimate of the local error near a tolerance, which hides the dependence we wish to expose.

All arithmetic is carried out by the MPFR library,[9] which rounds every operation correctly to a significand of p bits, with p chosen by the program. We call it from Python through the `gmpy2` package; Julia’s built-in `BigFloat` type wraps the same library and serves equally well. The values $p = 24, 53, 64$ and 113 reproduce the significand widths of IEEE single, double, extended and quadruple precision,[10] and $p = 237$, that of the 256-bit format, stands in for exact arithmetic: its rounding unit is $2^{-237} \approx 10^{-71}$.

Steps are powers of two, $h = 2^{-k}$. A power of two is represented exactly at every precision, so that lowering p changes the arithmetic and nothing else. A decimal step such as 10^{-4} would be rounded differently at each precision, and runs intended to differ only in their arithmetic would also be integrating with slightly different steps.

C. Reference solution and measure of error

The error of a run is measured by the distance $d(t)$ between its lower bob and that of a reference run, in units of l . Since both bobs lie within $2l$ of the pivot, d saturates near 4 once two runs have nothing to do with each other. We define the *horizon* T of a run as the first time at which $d \geq 10^{-2}$, which is roughly when two superimposed animations become visibly different.

No exact solution is available to serve as the reference, so the reference is simply a much more accurate run, and the following argument says how far it can be trusted. Let e_a and e_r be the true errors of a run and of the reference. By the triangle inequality the measured separation satisfies $|d - e_a| \leq e_r$. All errors are amplified at the same rate, so the ratio e_r/e_a stays close to its initial value, which for two RK4 runs with steps 2^{-k_r} and 2^{-k_a} is $2^{-4(k_r - k_a)}$. Against a reference with $k_r = 18$, the separation therefore gives the true error of a run with $k_a \leq 16$ to better than one part in 250, for as long as the run itself is accurate.

D. Three experiments

Truncation alone. With $p = 237$ we vary k from 4 to 17 and compare each run with the reference $k = 18$. Roundoff is some fifty orders of magnitude below the truncation error and plays no part.

Roundoff alone. With h fixed we vary p from 11 to 128 and compare each run with the run at $p = 237$ and the same h . Both runs iterate the same discrete map, so the truncation error is common to them and cancels exactly. What remains is the effect of rounding. This separation is the step that hardware arithmetic does not allow, and it is the reason for using a multiple-precision library.

Both together. With $p = 24, 32, 40, 53$ and 64 we vary k and compare with the reference of the first experiment. This is the situation of a student working in ordinary single or double precision.

III. THE HORIZON OF PREDICTABILITY

Let $\mathbf{e}_n$ be the error after n steps. One step maps it to $\mathbf{e}_{n+1} = \mathbf{J}_n \mathbf{e}_n + \boldsymbol{\tau}_n + \boldsymbol{\rho}_n$, where $\mathbf{J}_n$ is the Jacobian of the flow across the step, $\boldsymbol{\tau}_n = O(h^5)$ is the local truncation error of RK4, and

$\rho_n = O(2^{-p})$ is the rounding error committed in the step. Unrolling the recursion expresses the error at time t as a sum over all earlier steps of the error committed there, carried forward by the product of the intervening Jacobians. In a chaotic system that product grows on average as $e^{\lambda(t-t_j)}$, where λ is the largest Lyapunov exponent. The sum is then a geometric series dominated by its earliest terms, and

$$e(t) \approx (A h^4 + B 2^{-p} h^{-\gamma}) e^{\lambda t}. \quad (4)$$

The first term collects $1/h$ truncation errors of size h^5 per unit time. The second collects $1/h$ rounding errors of size 2^{-p} per unit time, which add like a random walk, giving $\gamma = \frac{1}{2}$, if their signs are independent and linearly, giving $\gamma = 1$, if they are not.[11] The constants A and B depend on the trajectory but not on h or p .

Setting $e(T) = \epsilon$ for a tolerance ϵ gives the horizon,

$$T = \frac{1}{\lambda} \ln \frac{\epsilon}{A h^4 + B 2^{-p} h^{-\gamma}}. \quad (5)$$

Everything in the exercise follows from Eq. (5). When truncation dominates and $h = 2^{-k}$,

$$T = T_A + \frac{4 \ln 2}{\lambda} k, \quad (6)$$

so that each halving of the step extends the horizon by the fixed amount $4 \ln 2 / \lambda$, the factor 4 being the order of the method. When roundoff dominates,

$$T = T_B + \frac{\ln 2}{\lambda} (p - \gamma k), \quad (7)$$

so that each additional bit extends the horizon by $\ln 2 / \lambda$, while each halving of the step now *shortens* it. The two regimes meet where the two terms of Eq. (4) are equal, at $k^* = [p + \log_2(A/B)] / (4 + \gamma)$, and the horizon there,

$$T_{\max}(p) \approx \text{const} + \frac{\ln 2}{\lambda} \frac{4p}{4 + \gamma}, \quad (8)$$

is the longest attainable with p bits and a fourth-order method. Both slopes are properties of the pendulum alone, which is what allows the numerical error to measure λ .

IV. RESULTS

A. Step size

Figure 1 shows what the experiment looks like. Three pendulums start from identical states and are integrated with identical 237-bit arithmetic. They differ only in h . For a

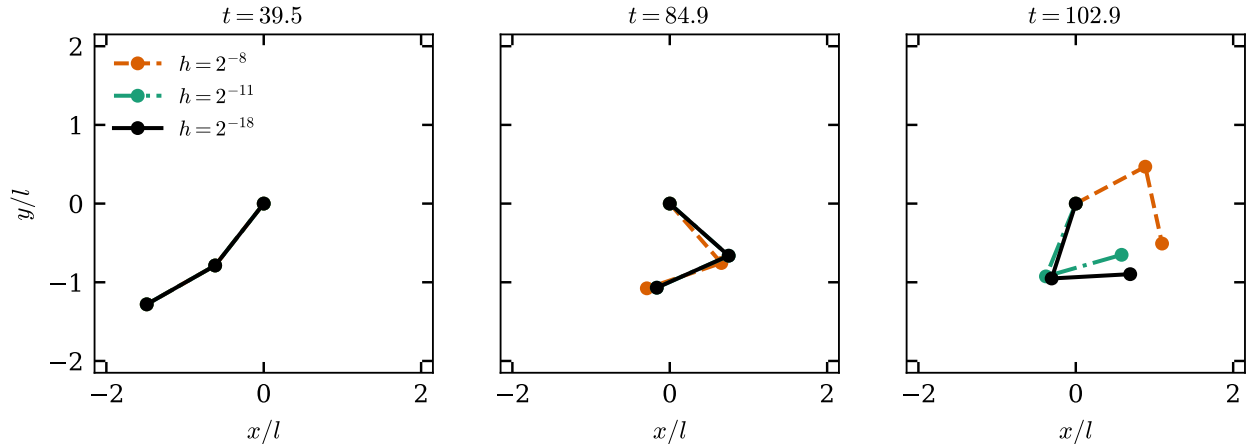


FIG. 1. Three double pendulums released from the same state and integrated with the same 237-bit arithmetic, differing only in the step h . The run with $h = 2^{-8}$ leaves the others near $t = 79$, and the run with $h = 2^{-11}$ near $t = 97$. Times are in units of $\sqrt{l/g}$.

long time they coincide to within the width of a line, and then the one with the largest step departs, followed later by the next. An animated version, in which the error curves are drawn alongside the pendulums as they evolve, is included in the supplementary material and makes the point to a class more quickly than any graph.

Figure 2(a) shows $d(t)$ for a selection of steps. Each curve rises through many orders of magnitude at the same average rate and then saturates. The curves are evenly spaced, and Fig. 2(b) shows why: multiplied by h^{-4} they collapse onto a single curve, to within 0.21 decades for every $h \leq 2^{-8}$ wherever none has saturated, and the error at $t = 10$ scales as $h^{4.000}$. That curve is the amplification factor of the trajectory itself. To confirm this we perturbed $\theta_1(0)$ by 2^{-170} , a perturbation small enough to remain in the linear regime for the entire run, and plotted the resulting separation as the thin line in Fig. 2(b). It follows the collapsed error curves through every plateau and burst. The error of the integrator grows exactly as a perturbation of the initial condition does, because it is one.

The growth is exponential only on average. The local rate fluctuates as the pendulum passes through more and less unstable parts of its motion, and those fluctuations are as much a part of the lesson as the mean.

The horizons are plotted against k in Fig. 3(a). A straight line fitted for $7 \leq k \leq 16$ has slope 7.9 ± 0.4 time units per halving of the step, and Eq. (6) converts that to $\lambda = 0.35 \pm 0.02$. The scatter about the line is not noise. It is the fluctuating local growth rate again: a horizon

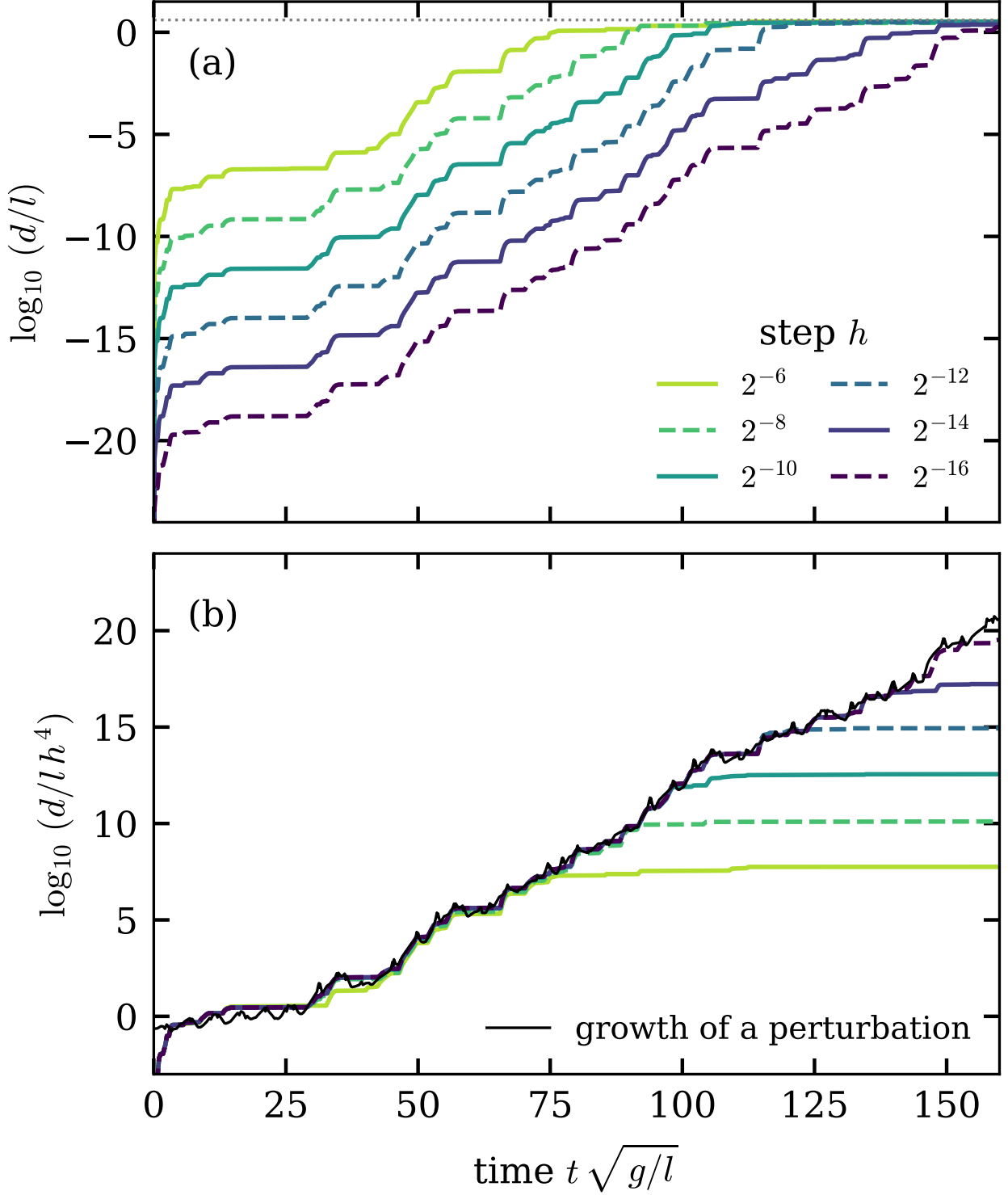


FIG. 2. (a) Separation d of the lower bob from that of the reference run ($h = 2^{-18}$) for six values of the step, in 237-bit arithmetic; the running maximum is shown. The dotted line marks the largest possible separation, $4l$. (b) The same curves multiplied by h^{-4} collapse. The thin black line is the separation of two runs whose initial conditions differ by 2^{-170} , shifted vertically.

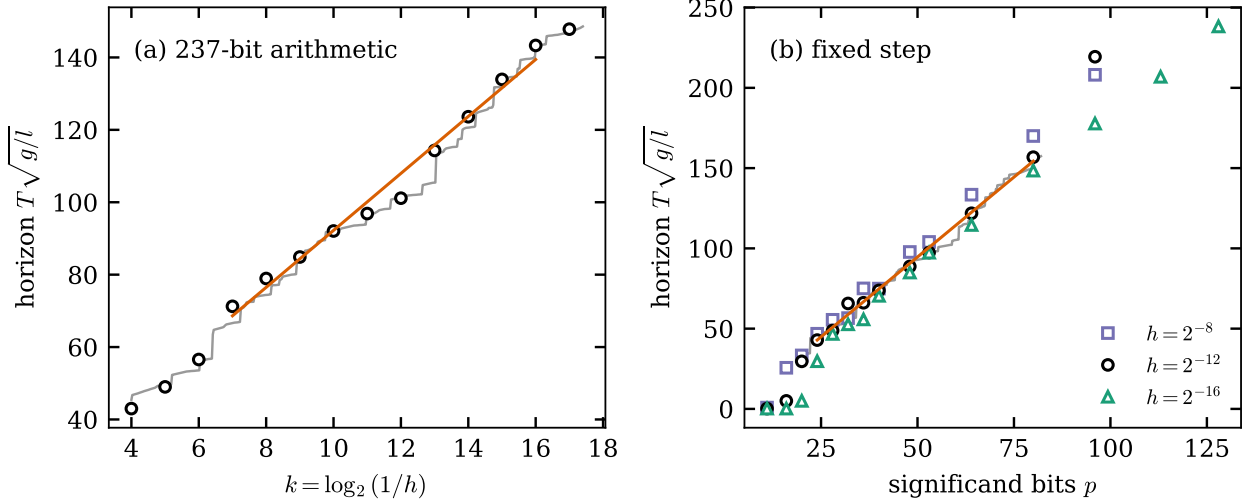


FIG. 3. Horizon T , the first time at which $d \geq 10^{-2}$. (a) Against $k = \log_2(1/h)$ in 237-bit arithmetic, where only truncation error is present. (b) Against the number of significant bits p at fixed step, where only roundoff is present. The straight lines are least-squares fits whose slopes give λ through Eqs. (6) and (7). The gray staircases are the horizons predicted from the measured growth of a perturbation of the initial condition.

that falls just before a burst of growth gains little from the next halving, and one that falls just after gains a great deal. The gray staircase in Fig. 3(a) makes this quantitative. It is the horizon predicted for every h from the measured growth of the 2^{-170} perturbation, with the single constant A of Eq. (4) taken from the collapse, and it passes through the measured horizons step for step.

B. Precision

Figure 4 shows the second experiment for $h = 2^{-12}$. The picture is the same with 2^{-p} in place of h^4 : evenly spaced curves, rising at a common rate. The horizons in Fig. 3(b) give a slope of 1.99 ± 0.06 time units per bit and hence, from Eq. (7), $\lambda = 0.35 \pm 0.01$. Changing the step by a factor of 256 moves the line only slightly. Comparing the errors at fixed early times for $h = 2^{-8}$ and 2^{-16} gives $\gamma \approx 0.6$ in Eq. (4), with values between 0.3 and 0.8 for individual precisions, which lies between the random-walk and the systematic limits.

The smallest precisions are excluded from the fit for a reason worth showing students. With $p = 11$ and $h = 2^{-12}$ the pendulum *never moves*, and the way it fails repays a minute's

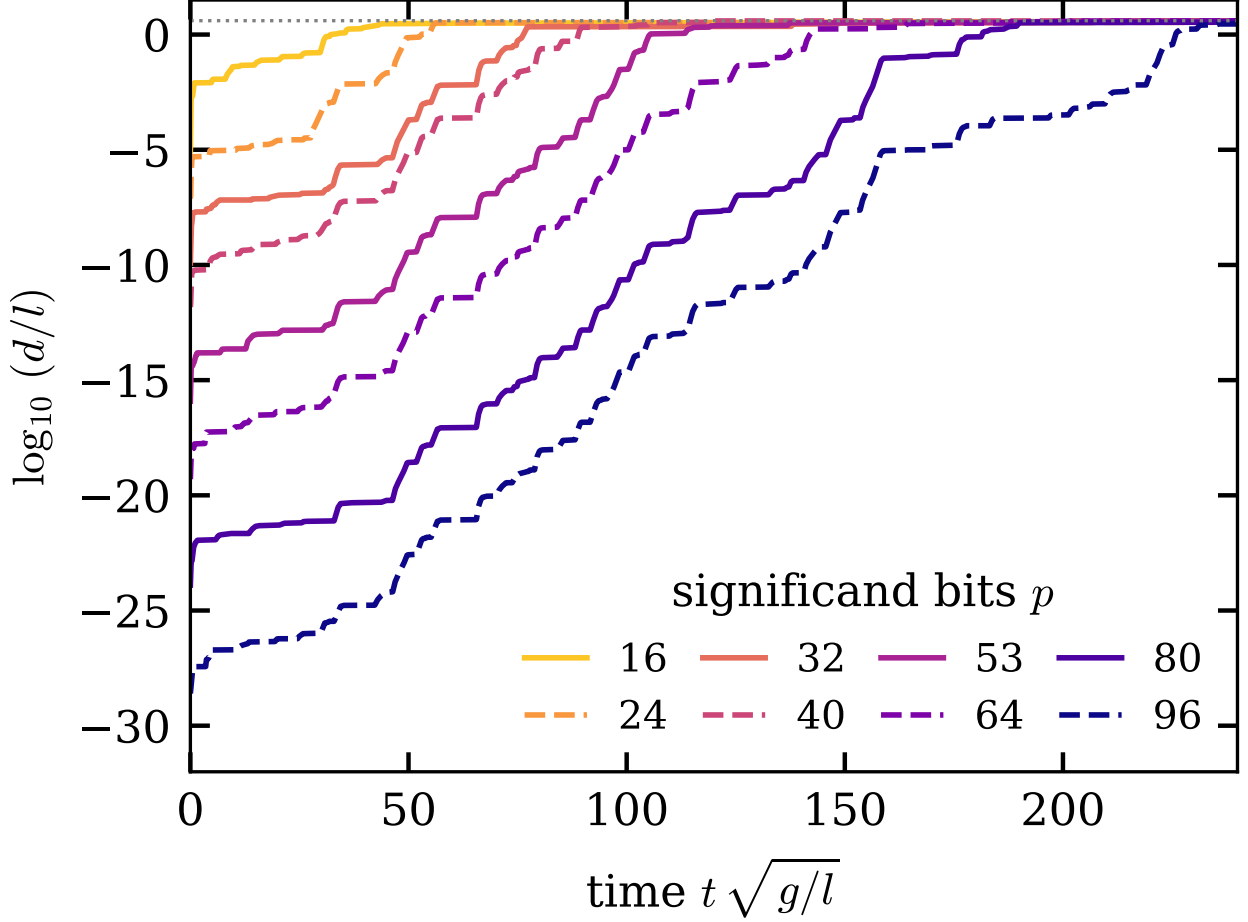


FIG. 4. Separation from the 237-bit run at the same step, $h = 2^{-12}$, for eight significand widths p . Truncation error is common to each pair of runs and cancels, so the curves show the effect of roundoff alone.

thought. The velocity does evolve at first: ω_1 falls by exactly h at every step for 2048 steps. At $t = \frac{1}{2}$ it reaches $-\frac{1}{2}$, where h is half a unit in the last place of ω_1 ; the sum is a tie, round-to-even returns $-\frac{1}{2}$, and the velocity is frozen. From then on the increment to the angle is $h|\omega_1| = 2^{-13}$, a quarter of the 2^{-11} needed to change the last of eleven bits in $\theta_1 = \pi/2$, and the computed pendulum hangs horizontal forever with a constant velocity that produces no motion. With the same eleven bits and $h = 2^{-8}$ the velocity would not freeze until $|\omega_1| = 8$, the increment $h|\omega_1|$ first exceeds 2^{-11} at $t = 0.129$, and the pendulum falls. The pattern is general for this release. The velocity freezes at 2^{p-k} and the angle needs $|\omega_1| > 2^{k-p}$, so the pendulum stalled for every $h < 2^{-p}$ that we tried and moved for every $h > 2^{-p}$; at $h = 2^{-p}$ the increment is itself a tie, and the outcome depends on the last bit

TABLE I. The largest Lyapunov exponent of the double pendulum released from rest with both rods horizontal, in units of $\sqrt{g/l}$. The last three rows sample the growth rate over the first 160 time units only, within which it averages 0.28 in the first half and 0.33 in the second. Uncertainties are standard errors of the fitted slopes. At $h = 2^{-8}$ and 2^{-16} the last row becomes 0.31 and 0.34.

Method	λ
Two-trajectory method, 2×10^4 time units	0.299
Growth of a perturbation of 2^{-170}	0.33
Horizon against step, Eq. (6)	0.35 ± 0.02
Horizon against precision, Eq. (7)	0.35 ± 0.01

of $\pi/2$. Here a smaller step has not reduced the error but made it total, and no analysis based on small perturbations describes what happens. It is an extreme case of the tendency expressed by the negative sign in Eq. (7).

C. Four measurements of the Lyapunov exponent

Table I collects the estimates. The reference value comes from the two-trajectory method of Benettin *et al.*: [12, 13] a shadow trajectory is started at a distance $d_0 = 10^{-8}$ from the fiducial one, and once per time unit the logarithm of the growth in their separation is accumulated and the shadow is pulled back to d_0 . Averaged over 2×10^4 time units this gives $\lambda = 0.299$ for our initial condition. Of 47 other initial conditions at rest on the same energy surface, 44 are chaotic, with a median of 0.296, and the remaining 3 lie on regular orbits with λ indistinguishable from zero. The direct measurement is the mean slope of the thin line in Fig. 2(b); it is 0.33 over the whole run and 0.35 for $t > 20$, once the slow start from rest is past.

The two estimates obtained from numerical error agree with each other and with the directly measured growth of a perturbation over the same interval. All three exceed the long-time value by some 18%. The difference is not an error of method. A horizon samples the growth rate only up to the horizon, and over its first 160 time units this trajectory happens to be more unstable than it is on average; the exponent proper is the limit of long times, and the finite-time values scatter about it. The distinction is worth making to

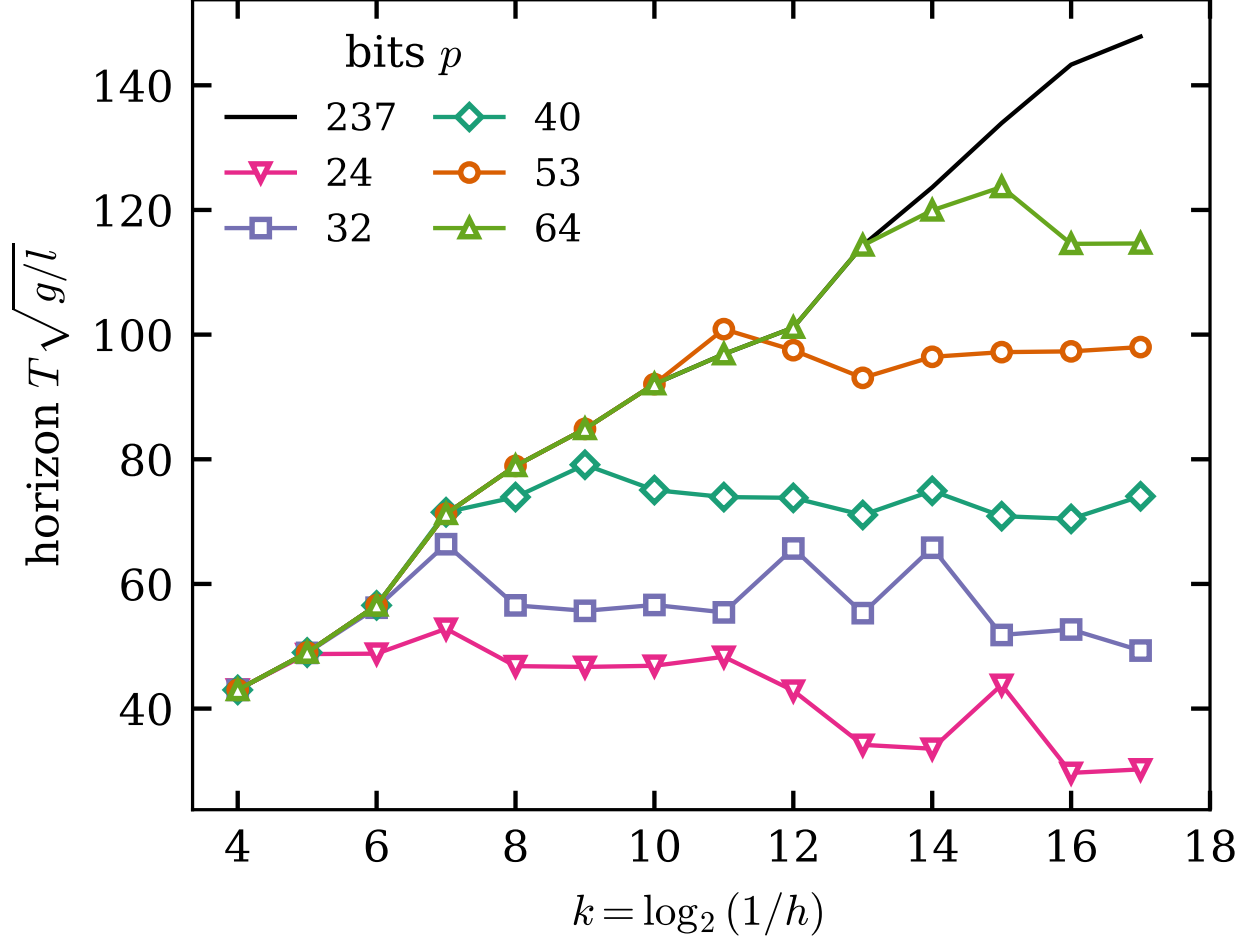


FIG. 5. Horizon against step for five working precisions, with the 237-bit result of Fig. 3(a) for comparison. For each precision there is a step beyond which further refinement does not help.

students, who otherwise picture exponential divergence as uniform. We regard the agreement as the central result of the exercise: a quantity students have been taught to regard as a defect of the computation has measured a property of the physical system.

D. Working precision

Figure 5 shows the third experiment. At large steps every precision gives the same horizon, because truncation dominates and the arithmetic is irrelevant. As the step shrinks each curve leaves the 237-bit line in turn and levels off, and at the lower precisions it then declines, as Eq. (5) predicts. Table II gives the best step and the longest horizon for each format.

TABLE II. Longest horizon attainable with RK4 at each precision, and the step at which it occurs. The last column is for rods of length 1 m.

Significand bits p	Best step	$T_{\max}\sqrt{g/l}$	$T_{\max}$ (s)
24 (single)	2^{-7}	53	17
32	2^{-7}	66	21
40	2^{-9}	79	25
53 (double)	2^{-11}	101	32
64 (extended)	2^{-15}	124	40

In double precision the horizon is about 101 time units, or 32 s for a pendulum with one-meter rods, and it is reached at $h = 2^{-11}$. A student who responds to a doubtful result by reducing the step to 2^{-17} does 64 times the work and gains nothing: the horizon is 98. In single precision the horizon is 53 time units, reached already at $h = 2^{-7}$, and by $h = 2^{-17}$ it has fallen to 30.

E. Energy

Figure 6 compares the two errors for an ordinary double-precision run at $h = 2^{-10}$. The error in the energy grows slowly, through the secular drift of a non-symplectic method, and at the end of the run is still below 10^{-10} . The error in the position reaches the size of the apparatus at $t \approx 92$. The reason is geometric. Energy constrains the state to a three-dimensional surface in the four-dimensional phase space, and the divergence of nearby trajectories takes place within that surface. A computed state can drift an arbitrary distance along it without leaving it.

V. DISCUSSION

A. What accuracy costs

The work of a fixed-step integration is proportional to 2^k . By Eq. (6), extending the horizon by ΔT requires $\Delta k = \lambda \Delta T / (4 \ln 2)$, so the work is multiplied by $e^{\lambda \Delta T / 4}$; to stay on the optimum of Eq. (8) the precision must rise as well, by about $\lambda \Delta T (4 + \gamma) / (4 \ln 2)$

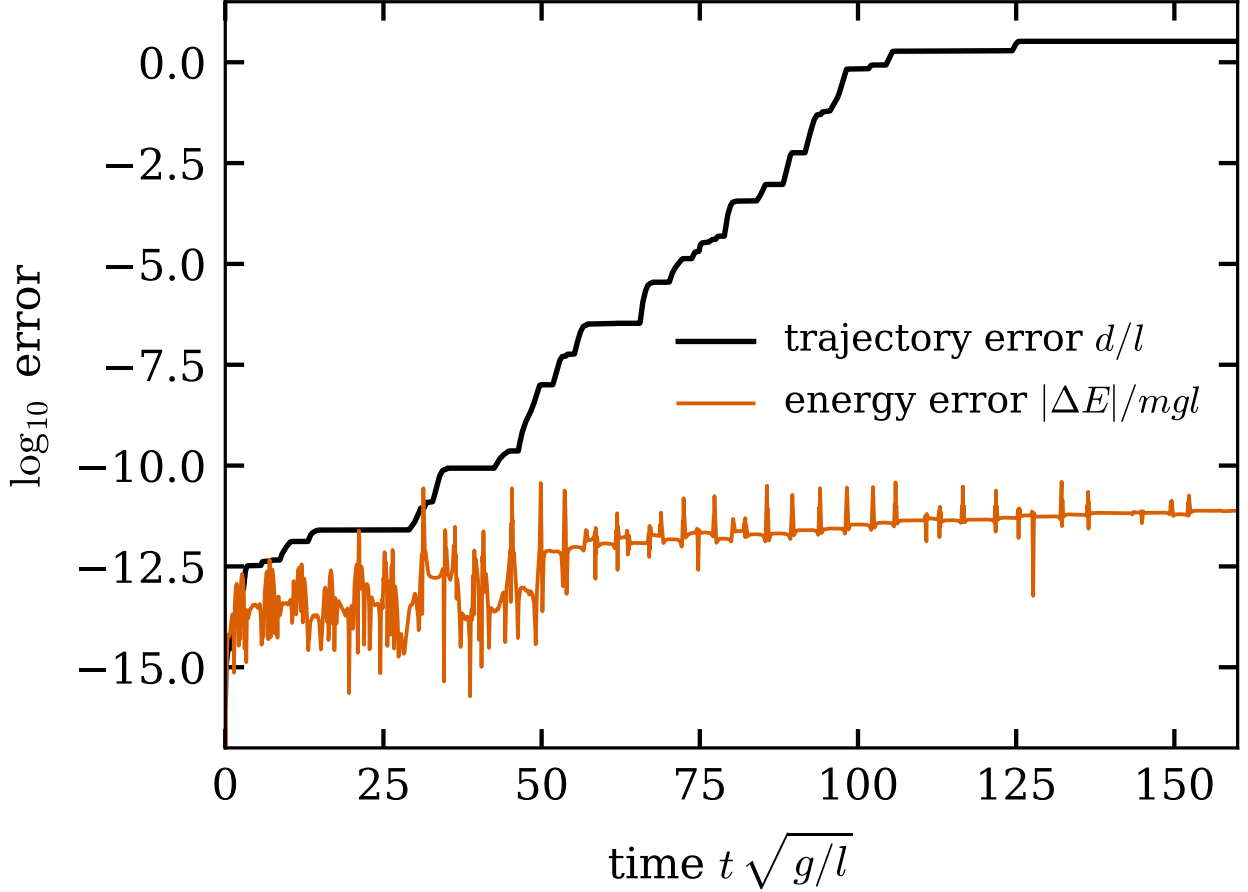


FIG. 6. Error in the position of the lower bob and error in the energy for one run in double precision ($p = 53$) with $h = 2^{-10}$. Energy remains conserved to better than a part in 10^{10} throughout, while the trajectory is lost near $t = 92$.

bits. Which λ to use depends on the interval in question. Doubling the horizon attainable in double precision means extrapolating beyond the times sampled here, for which the long-time exponent 0.299 is the appropriate rate: the number of steps is multiplied by roughly $e^{\lambda T_{\max}/4} \approx 2000$ and some 50 additional bits are called for. At the finite-time rate 0.35 measured over the first 160 time units the figures would be nearer 7000 and 59. Either way the horizon grows as the logarithm of the effort spent on it. A method of order n replaces the 4 by n , which is the real argument for high-order integrators in this setting. We checked the low end. At a fixed early time the error of forward Euler scales as $h^{1.002}$, that of the explicit midpoint rule as $h^{1.999}$, and that of RK4 as $h^{4.000}$, and the horizons lengthen in proportion: forward Euler with $h = 2^{-16}$ is accurate for 31 time units, which is less than the 43 that RK4 achieves with $h = 2^{-4}$ and a thousandth of the work.

In everyday units the result is easy to remember. For rods one meter long the long-time exponent is $\lambda = 0.94 \text{ s}^{-1}$, so reducing the step tenfold buys $4 \ln 10 / \lambda \approx 10 \text{ s}$ of accurate motion, or 8 s at the finite-time rate, at ten times the cost.

B. What is still right after the horizon

Beyond T the computed trajectory is not the solution of the initial-value problem that was posed. It is not meaningless, however. Shadowing results[14–17] show that, for long times, a noisy computed orbit of a chaotic system stays close to *some* exact orbit with a slightly different initial condition. Quantities that are properties of the ensemble of orbits rather than of a particular one, such as the Lyapunov exponent, are therefore computed correctly by runs hundreds of horizons long. The first row of Table I is such a computation, done in double precision. The distinction between predicting a trajectory and sampling the dynamics is the right note on which to leave a class.

C. Use in a course

The exercise suits a course in computational physics, classical mechanics or numerical analysis at the upper-division level, after students have met RK4 and floating-point numbers. The program is short, since the integrator is the usual one with the number type changed, and every run in this paper completes in minutes on a laptop except the two finest references. Students who have only hardware arithmetic can still do the third experiment, using a double-precision run at the best step of Table II as the reference for single precision. The original version of this project was carried out in exactly that spirit, with decimal steps and a handful of precisions, and the logarithmic law was evident even there: each tenfold reduction of the step bought between five and ten seconds.

VI. SUGGESTED PROBLEMS

1. *Order of the method.* Replace RK4 with forward Euler and with the midpoint rule, repeat the first experiment, and show that the slope of T against k is $n \ln 2 / \lambda$ for a method of order n . Estimate the step that Euler’s method would need to match RK4

at $h = 2^{-10}$.

2. *Energy dependence.* Release the pendulum from rest at smaller angles. Show that λ decreases and the horizon lengthens, and that for sufficiently small angles the error grows as a power of t rather than exponentially. What does Eq. (5) become in that case?
3. *The roundoff exponent.* Repeat the second experiment at several steps and determine γ . Then replace the update $\mathbf{y} \leftarrow \mathbf{y} + \Delta\mathbf{y}$ by compensated summation[18] and determine it again.
4. *A symplectic method.* Integrate with the implicit midpoint rule, which is symplectic. Show that the energy error no longer drifts, and compare the horizon with that of the explicit midpoint rule at the same step. Does preserving the geometry of phase space help with predicting the trajectory?
5. *Another system.* Carry out the first and third experiments for the Lorenz equations with the standard parameters, for which $\lambda \approx 0.906$, and predict the double-precision horizon before measuring it.
6. *Shadowing.* Compute the time average of the kinetic energy over 10^4 time units in single and in double precision with $h = 2^{-7}$. We find 1.357 and 1.358 in units of mgl . Explain why the two agree to a part in a thousand although the trajectories separate after fifty time units.

SUPPLEMENTARY MATERIAL

The supplementary material contains the Python source for every computation and figure in this paper, the numerical values plotted, and two animations: pendulums differing only in step size, and pendulums differing only in precision, each drawn beside its growing error.

AUTHOR DECLARATIONS

The authors have no conflicts to disclose.

- [1] T. Shinbrot, C. Grebogi, J. Wisdom, and J. A. Yorke, Chaos in a double pendulum, *Am. J. Phys.* **60**, 491 (1992).
- [2] R. B. Levien and S. M. Tan, Double pendulum: An experiment in chaos, *Am. J. Phys.* **61**, 1038 (1993).
- [3] S. D’Alessio, An analytical, numerical and experimental study of the double pendulum, *Eur. J. Phys.* **44**, 015002 (2023).
- [4] E. N. Lorenz, Deterministic nonperiodic flow, *J. Atmos. Sci.* **20**, 130 (1963).
- [5] J. Lighthill, The recently recognized failure of predictability in Newtonian dynamics, *Proc. R. Soc. Lond. A* **407**, 35 (1986).
- [6] E. Hairer, S. P. Nørsett, and G. Wanner, *Solving Ordinary Differential Equations I: Nonstiff Problems*, 2nd ed., Springer Series in Computational Mathematics, Vol. 8 (Springer, Berlin, 1993).
- [7] J. Teixeira, C. A. Reynolds, and K. Judd, Time step sensitivity of nonlinear atmospheric models: Numerical convergence, truncation error growth, and ensemble design, *J. Atmos. Sci.* **64**, 175 (2007).
- [8] S. Liao, On the reliability of computed chaotic solutions of non-linear differential equations, *Tellus A* **61**, 550 (2009).
- [9] L. Fousse, G. Hanrot, V. Lefèvre, P. Pélicier, and P. Zimmermann, MPFR: A multiple-precision binary floating-point library with correct rounding, *ACM Trans. Math. Softw.* **33**, 13 (2007).
- [10] D. Goldberg, What every computer scientist should know about floating-point arithmetic, *ACM Comput. Surv.* **23**, 5 (1991).
- [11] E. Hairer, R. I. McLachlan, and A. Razakarivony, Achieving Brouwer’s law with implicit Runge–Kutta methods, *BIT Numer. Math.* **48**, 231 (2008).
- [12] G. Benettin, L. Galgani, A. Giorgilli, and J.-M. Strelcyn, Lyapunov characteristic exponents for smooth dynamical systems and for Hamiltonian systems; a method for computing all of

- them. Part 1: Theory, *Meccanica* **15**, 9 (1980).
- [13] A. Wolf, J. B. Swift, H. L. Swinney, and J. A. Vastano, Determining Lyapunov exponents from a time series, *Physica D* **16**, 285 (1985).
 - [14] S. M. Hammel, J. A. Yorke, and C. Grebogi, Do numerical orbits of chaotic dynamical processes represent true orbits?, *J. Complexity* **3**, 136 (1987).
 - [15] S. M. Hammel, J. A. Yorke, and C. Grebogi, Numerical orbits of chaotic processes represent true orbits, *Bull. Am. Math. Soc.* **19**, 465 (1988).
 - [16] C. Grebogi, S. M. Hammel, J. A. Yorke, and T. Sauer, Shadowing of physical trajectories in chaotic dynamics: Containment and refinement, *Phys. Rev. Lett.* **65**, 1527 (1990).
 - [17] T. Sauer, C. Grebogi, and J. A. Yorke, How long do numerical chaotic solutions remain valid?, *Phys. Rev. Lett.* **79**, 59 (1997).
 - [18] N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, 2nd ed. (SIAM, Philadelphia, 2002).